

Simulation of Si-Pixel vx- μ SR using musrSim

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Abstract

This report documents a simplified **musrSim**/Geant4 study of a Si-pixel vertex-reconstructed μ SR geometry inspired by Mandok et al. (2026). The purpose is not to reproduce the full mechanical detector module, but to test the essential ingredients of the reconstruction chain: transport of surface muons through thin silicon tracking layers, reconstruction of the incoming muon trajectory from upstream hits, extrapolation to the target plane, comparison with the true muon stopping position, and, in a second simulation, the reconstruction of a transverse-field vx- μ SR time spectrum from decay positron tracks. Two configurations are considered: a zero-field track-reconstruction study with an aluminium target and variable inner-layer spacing, and a transverse-field vx- μ SR study using a smaller aluminium target in a uniform magnetic field of 6.3 mT.

1 Simulation Geometry and Configuration

The simulated geometry is a simplified version of the Si-pixel vx- μ SR spectrometer described by Mandok et al. (2026). The beam axis defines the z -direction, the vertical detector coordinate is y , and the target is centered at $z = 0$. The incoming particles are positive surface muons traveling along $+z$ -direction, with initial spin polarization along $+x$ -direction.

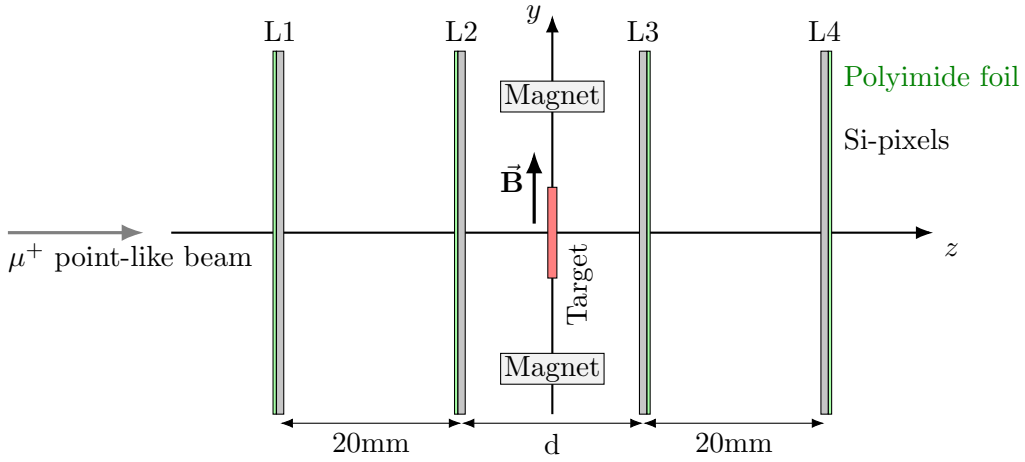


Figure 1: Schematic yz -cross section of the simplified Si-pixel vx- μ SR geometry. The z -axis is the beam direction and the aluminium target is centered at $z = 0$. Layers L1–L2 reconstruct the incoming muon track, while L3–L4 are used for downstream positron tracking. In the transverse-field configuration, a uniform field is applied along $+y$. The drawing is not to scale.

The detector is approximated by four continuous silicon tracking planes rather than by the full Quad-module structure. Each silicon layer is followed by a thin polyimide/Kapton support foil. The detector-layer positions used for the reference geometry are summarized in Table 1.

The two iron magnet volumes are included only as simple geometrical placeholders. In the transverse-field simulation the relevant field is not produced by a realistic magnet model, but is

Table 1: Tracking-layer geometry used in the simplified musrSim model.

Layer	Position z [mm]	Role	Detector ID
L1	-30	upstream outer layer	101
L2	-10	upstream inner layer	102
L3	+10	downstream inner layer	103
L4	+30	downstream outer layer	104

Table 2: Main detector, target, beam, and field parameters used in the simulations.

Component	Parameter	Value / implementation
Si tracking layers	active area	40 mm $\times$ 40 mm
	thickness	100 μ m
	material	G4_Si
Support foils	thickness	25 μ m
	material	G4_KAPTON
Target, Simulation 1	material and size	G4_Al, $D = 10$ mm, $H = 1$ mm
Target, Simulation 2	material and size	G4_Al, $D = 6$ mm, $H = 1$ mm
Primary beam	particle	μ^+
	kinetic energy	4.1 MeV
	direction	$+z$
	initial polarization	$+x$, polarization fraction = 100%
Beam model	transverse profile	point-like, monoenergetic beam; no collimator
Magnetic field	Simulation 1	$B = 0$
	Simulation 2	uniform field, $\vec{B} = (0, 6.3, 0)$ mT
Magnet placeholders	material and size	G4_Fe, 10 mm $\times$ 2 mm $\times$ 10 mm

imposed as a uniform field in the target region. The field points along $+y$, perpendicular to both the beam direction and the initial muon spin.

The electromagnetic physics configuration includes multiple scattering, ionisation, bremsstrahlung, and pair-production processes for positive muons, the corresponding electromagnetic processes for electrons and positrons, positron annihilation, and the standard photon interactions. A user step limit of 0.01 mm is applied in the thin silicon layers and in the target in order to improve tracking through small volumes.

Software Environment

For reproducibility, all simulation macros and analysis scripts are archived in the project github repository. The simulations were performed with `musrSim` 1.0.5 using Geant4 10.3, and the ROOT output was analysed with ROOT 6.36.06, `uproot` 5.7.4, and Python 3.12.13.

2 Simulation 1: Track-Reconstruction Residual versus Inner-Layer Distance

The first simulation campaign quantifies how the track-reconstruction residual depends on the distance d between the two inner tracking layers L2 and L3. For each detector configuration, 10^5 primary μ^+ events were simulated. The magnetic field was set to zero, and the target was an aluminium disk centered at $z = 0$ with diameter $D = 20$ mm and thickness $H = 1$ mm.

For each accepted event, the incoming muon trajectory is reconstructed from the hit positions in the two upstream layers L1 and L2. A straight-line tracklet is formed and extrapolated to

Table 3: Simulation conditions for the reconstruction-residual study.

Quantity	Value / selection
Primary particle	μ^+
Muon kinetic energy	4.1 MeV
Beam direction	$+z$
Initial polarization	$+x$
Magnetic field	$B = 0$
Target	G4.A1, $D = 20$ mm, $H = 1$ mm
Inner-layer distance	$d = 5$ mm, 10 mm, $\dots$, 40 mm
Events per configuration	10^5

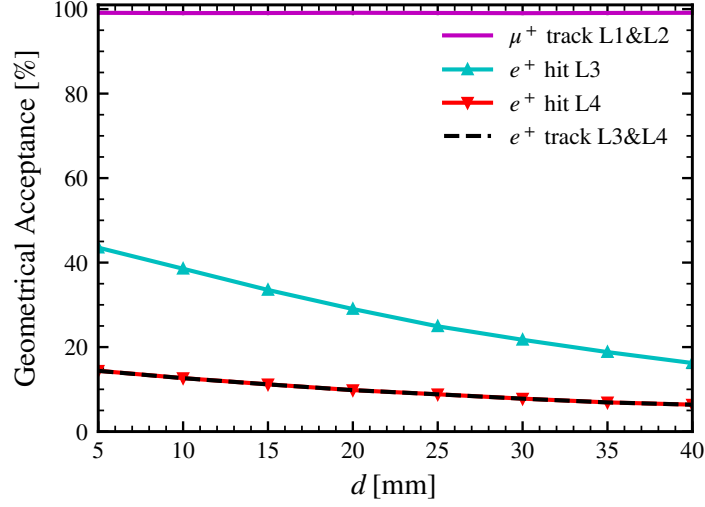


Figure 2: Simulation-level geometrical acceptance as a function of the inner-layer distance d . The μ^+ track acceptance is defined by the presence of valid hits in both upstream layers L1 and L2. The positron single-layer acceptances are defined by the presence of a hit in L3 or L4, normalized to the number of muons decaying in the target. The e^+ track acceptance requires valid hits in both L3 and L4 normalized to the number of muons decaying in the target. No detector-response inefficiency is included; therefore the curves represent geometrical and analysis-selection acceptance, not experimental detector efficiency.

the target plane. The extrapolated transverse position is then compared with the true simulated muon stopping position in the target. The simulation-level geometrical acceptance as a function of the distance d for both muon and positrons are shown in Fig. 2.

The muon reconstruction residual is defined as

$$\delta_\mu = \sqrt{(x_{\text{stop}} - x_{\text{ext}})^2 + (y_{\text{stop}} - y_{\text{ext}})^2},$$

where $(x_{\text{stop}}, y_{\text{stop}})$ is the true muon stopping position and $(x_{\text{ext}}, y_{\text{ext}})$ is the extrapolated position obtained from the L1–L2 tracklet. Only events with a valid target stop and a clean upstream muon track candidate are used for the δ_μ distribution.

An analogous residual is computed for decay positrons. In this case, positron tracklets are reconstructed from the downstream layers L3 and L4 and extrapolated backwards to the true muon decay position in the target. The positron residual is

$$\delta_{e^+} = \sqrt{(x_{\text{dec}} - x_{\text{ext}})^2 + (y_{\text{dec}} - y_{\text{ext}})^2}.$$

For each value of d , the distributions of δ_μ and δ_{e^+} are summarized by their mean values,

$$\langle\delta_\mu\rangle, \quad \langle\delta_{e+}\rangle,$$

and standard deviations,

$$\sigma(\delta_\mu), \quad \sigma(\delta_{e+}).$$

The standard deviation $\sigma(\delta_\mu)$ is used as the main estimate of the lateral uncertainty of the reconstructed muon stopping position. The distribution of δ_μ and δ_e for a fixed d is shown in Fig. 5 in the Appendix. The dependence of the residual mean and width on the inner-layer separation d is shown in Fig. 3.

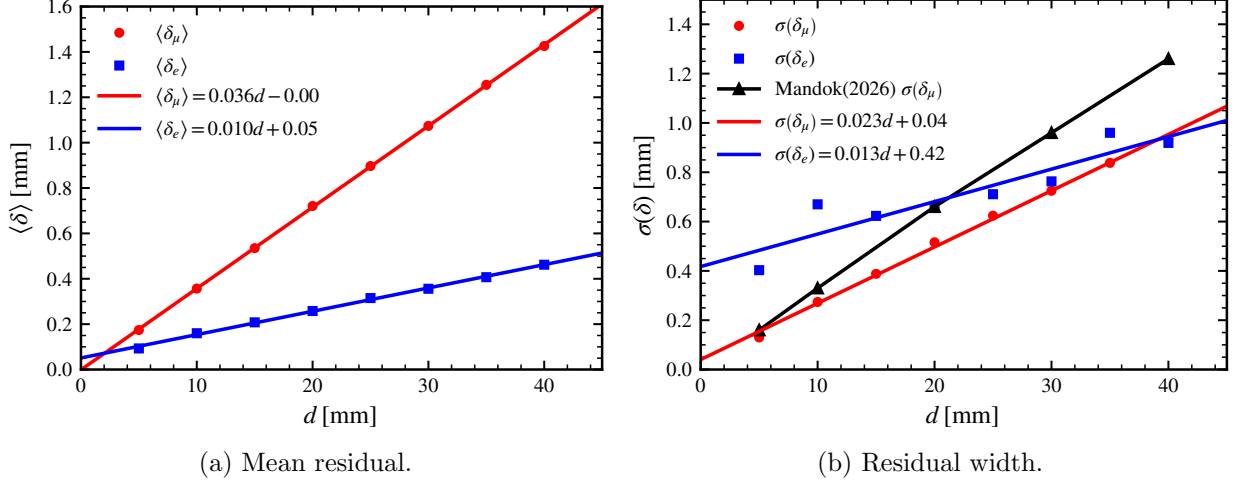


Figure 3: Mean and standard deviation of the reconstruction residuals as functions of the inner-layer distance d . The width $\sigma(\delta_\mu)$ is used as the main estimate of the muon vertex-reconstruction uncertainty.

3 Simulation 2: Transverse-Field vx- μ SR Spectrum

The second simulation tests whether the simplified geometry can reproduce the basic ingredients of a transverse-field vx- μ SR measurement. The detector geometry is kept fixed at the reference configuration with an inner-layer separation $d = 20$ mm. The target is changed to a smaller aluminium disk centered at $z = 0$, with diameter $D = 6$ mm and thickness $H = 1$ mm. A uniform magnetic field is applied in the target region,

$$\vec{B} = (0, 6.3 \times 10^{-3} \text{ T}, 0),$$

corresponding to a transverse field of $B = 6.3$ mT along $+y$. Since the beam travels along $+z$ and the initial muon-spin polarization is along $+x$, the spin precesses around the magnetic-field direction.

Two spectra are constructed. The first is a truth-level spectrum, where the muon decay time and positron direction are taken directly from simulated quantities. The second is a reconstructed vx- μ SR spectrum. In the reconstructed case, the incoming muon track is reconstructed from L1–L2 hits, while decay-positron tracklets are reconstructed using either the upstream or downstream detector pairs. Events are accepted if the muon decays in the target, the required muon and positron tracklets are present, the reconstructed positron–muon matching distance satisfies

$$d_{\text{match}} \leq 1 \text{ mm},$$

and the reconstructed time difference lies inside the analysis window,

Table 4: Simulation conditions for the transverse-field vx- μ SR study.

Quantity	Value / implementation
Primary particle	μ^+
Muon kinetic energy	4.1 MeV
Target	G4_A1, $D = 6$ mm, $H = 1$ mm
Inner-layer distance	$d = 20$ mm
Magnetic field	$\vec{B} = (0, 6.3 \times 10^{-3} \text{ T}, 0)$
Field direction	$+y$
Initial spin direction	$+x$
Time window	$0 < \Delta t < 13 \mu\text{s}$
Events simulated	10^5

$$0 < \Delta t < 13 \mu\text{s}.$$

The reconstructed time differences are defined as

$$\Delta t_{\text{up}} = t_{e^+, \text{up}}^{\text{rec}} - t_{\mu}^{\text{rec}}, \quad \Delta t_{\text{down}} = t_{e^+, \text{down}}^{\text{rec}} - t_{\mu}^{\text{rec}}.$$

The upstream and downstream spectra are then fitted with a muon-decay envelope modulated by transverse-field spin precession,

$$N(t) = N_0 e^{-t/\tau_{\mu}} [1 + A \cos(\omega t + \phi)], \quad \tau_{\mu} = 2.197 \mu\text{s}.$$

The asymmetry is computed from the two spectra as

$$A(t) = \frac{N_{\text{up}}(t) - \alpha N_{\text{down}}(t)}{N_{\text{up}}(t) + \alpha N_{\text{down}}(t)}, \quad \alpha = \frac{\sum_t N_{\text{up}}(t)}{\sum_t N_{\text{down}}(t)}.$$

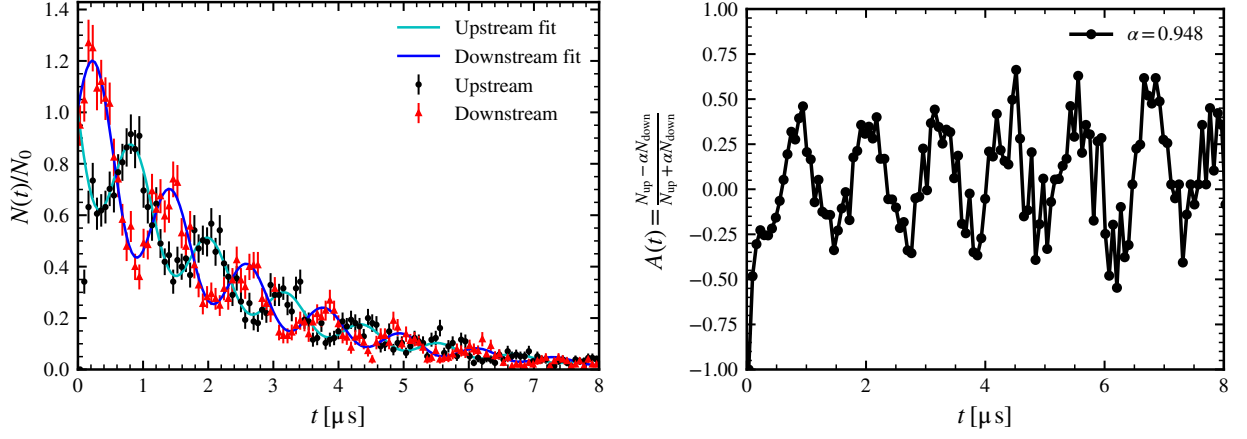
The resulting spectra and asymmetry are shown in Fig. 4. The purpose of this simulation is not yet a precision extraction of the relaxation parameters, but a consistency check that the simulated detector geometry, event matching, and time reconstruction produce the expected transverse-field oscillatory structure.

4 Current Simplifications

The present model keeps only the ingredients needed for a first reconstruction and vx- μ SR feasibility test. The real Quad-module geometry, individual chip gaps, pixelization, PCB material, detailed support structures, realistic beam spot, Pb collimation, detector timing response, time-over-threshold response, and realistic permanent-magnet field map are not included. The magnetic volumes in the macros are placeholders; the field used in the second study is a uniform field applied to the target region.

5 Analysis Outputs

The reconstruction-resolution study produces residual histograms for μ^+ and e^+ , the mean and standard deviation of the residuals as functions of d , and basic layer-efficiency checks. The vx- μ SR study produces target stopping plots, ideal and reconstructed time spectra, fitted oscillatory spectra, and upstream-downstream asymmetry plots. These outputs provide a compact validation chain from geometrical tracking to a reconstructed transverse-field vx- μ SR signal.



(a) Reconstructed upstream and downstream spectra.

(b) Corresponding upstream-downstream asymmetry.

Figure 4: Reconstructed transverse-field vx- μSR signal for a 6 mm-diameter aluminium target in a uniform 6.3 mT field. The spectra are built from matched muon and positron tracklets inside the time window $0 < \Delta t < 13 \mu\text{s}$. The asymmetry shows the expected oscillatory behaviour, although the present statistics are limited.

A Appendix: Sanity Checks and Supplementary Plots

A.1 Simulation 1

A.2 Simulation 2

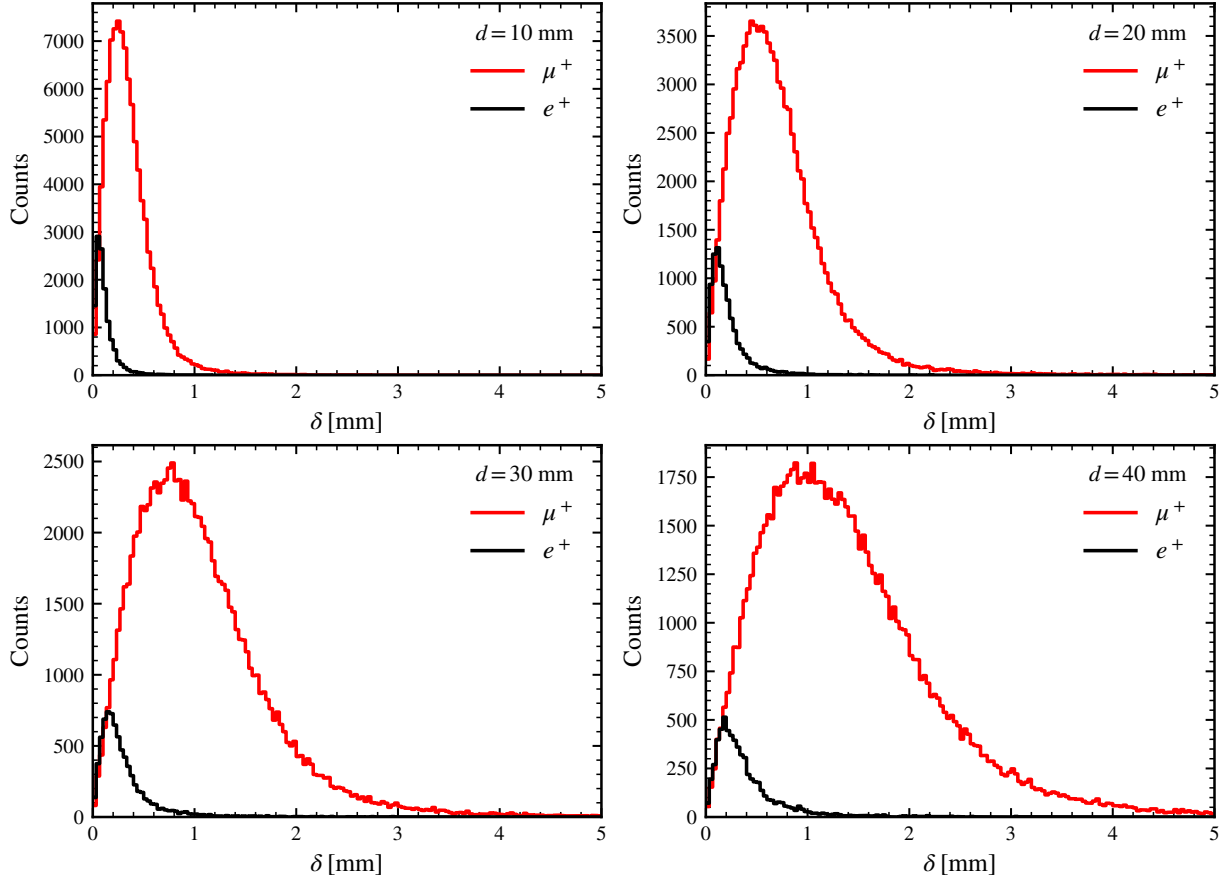


Figure 5: Distribution of δ_μ (red) and δ_e (black) for different distances d .

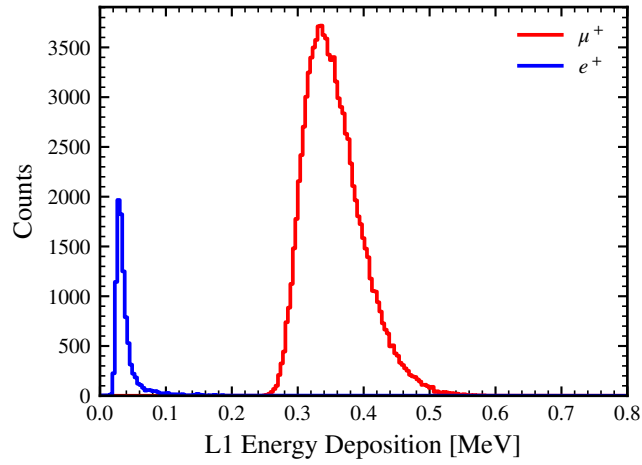


Figure 6: Energy deposit from muons and positrons in the first $L1$ upstream layer.

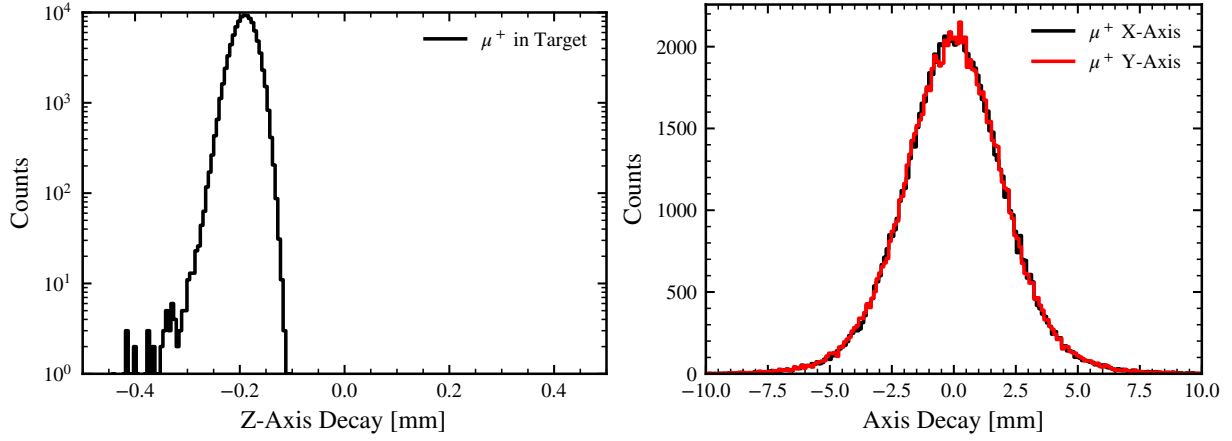


Figure 7: Muon stop and decay in the target (Al disk with diameter 20mm and thickness 1mm). Panel on the right shows muon stop along the z direction (depth) whereas, panel on the right shows the transverse spread of muons in the target.

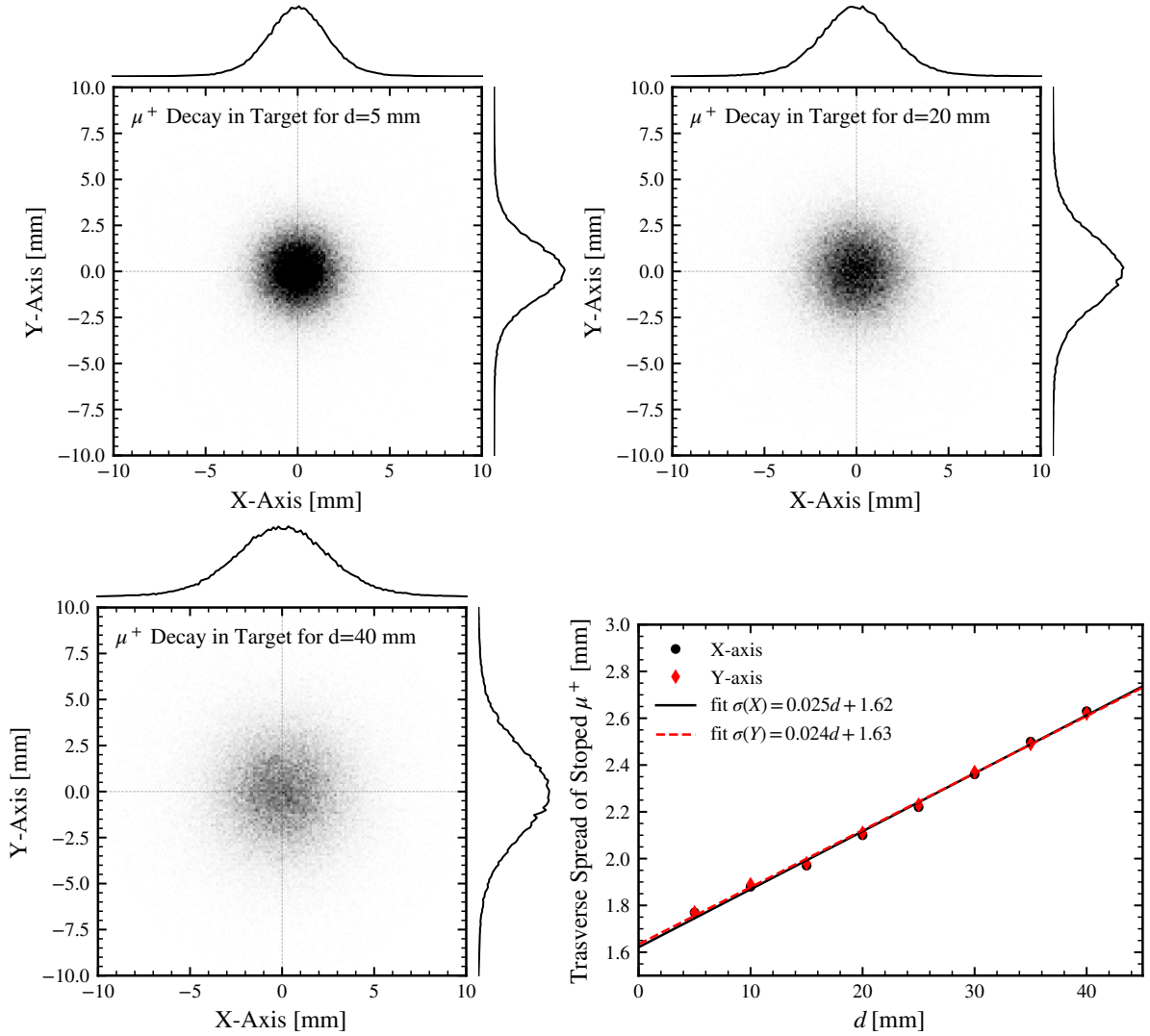


Figure 8: Top and lower left: Transverse spread of muon beam in the target for $d = 5\text{mm}$ (top left), $d = 20\text{mm}$ (top right) and $d = 40\text{mm}$ (bottom left). Bottom right: Dependence of the standard deviation (width) of the transverse beam spread in the target as a function of d between L2 and L3. The Dependence is linear and the fit function are shown in the figure.

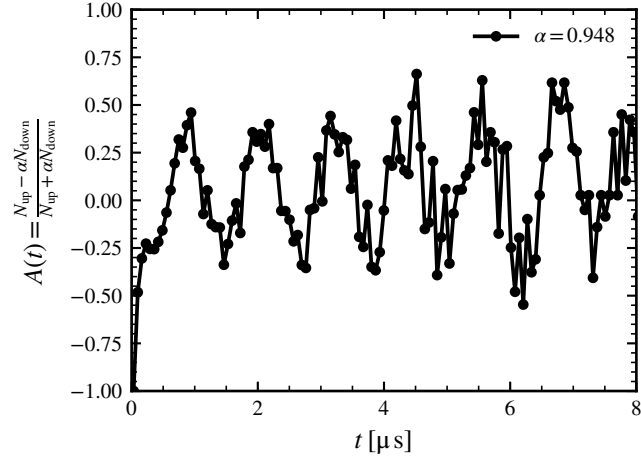


Figure 9: Asymmetry parameter for the vx- μ SR spectrum.